

Skills Enhancement Web Application

Platform Objective: The platform aims to provide a secure, interactive, and personalized cloud learning ecosystem for students, professionals, and corporate users to gain hands-on experience, certification readiness, and career assistance.

- I. **Latest Offerings:** A rolling banner showcasing new courses and features.
- II. **Guided Lab Exercises:** Step-by-step learning paths for hands-on cloud skill development.
- III. **User Registration:** A section for new users to sign up.
- IV. **User Dashboard:** A personalized dashboard for users to opt into services, track progress, and manage their learning journey.
- V. **Success Stories & Testimonials:** Featuring user achievements.
- VI. **Certification Test Series:** Practice exams for cloud certifications.
- VII. **Other pages (Home Page):**
 - a. About Us
 - b. Why Choose Us
 - c. Support section with options to send an email or call, along with a FAQ section for quick answers to common queries.
 - d. Auto email section for inquiry
 - e. Integration with WhatsApp (*Users can message your WhatsApp number and get responses*)
 - f. Platform should support adding pages
- VIII. **Cloud Account Services:** Option to purchase cloud-based services with flexible duration plans.
- IX. **Subscription & Discounts:** Purchase training modules with exclusive discounts and free coupons.
- X. **Instructor Resources:** Trainers can download course materials.
- XI. **Instructor-Led Training:** Scheduled live training sessions.
- XII. **On-Demand Video Training:** Pre-recorded video sessions for flexible learning.
- XIII. **Career Assistance:** Resume-building support and interview preparation.
- XIV. **Exam Voucher:** Option to buy certification exam vouchers.
- XV. **Skill Assessment Quiz:** Quiz to recommend personalized learning paths.
- XVI. **Free Webinars & Training:** An option to register for upcoming free sessions, displayed dynamically.
- XVII. **Live Events & Expert Talks:** A schedule of industry events, expert discussions, and training sessions.
- XVIII. **Multi-Currency Payments:** Support for both INR and USD transactions.
- XIX. **Analytics Integration:** Connect with free analytics platforms like Google Analytics, Microsoft Clarity or any other analytical tool for performance insights.
- XX. **Additional Pages:** Flexible CMS structure allowing creation of new pages as the platform evolves.

Security Requirement

- I. Copying content directly from the webpage should be restricted.
- II. Users cannot log in to the same account from multiple devices simultaneously
 - a. or invalidate the previous session when a new login is detected.
- III. Downloading or saving images and videos from the platform should not be permitted.
- IV. Automatic logout after a period of inactivity to enhance security.
- V. Restrict direct access to website content via sub-links.
- VI. Encrypt sensitive user data in transit and at rest.
- VII. Apply dynamic watermarks with user details on lab environments, images and videos.
- VIII. Prevent cached by browsers.
- IX. Maintain logs of user activities.
- X. Enable OTP-based registration and enforce Multi-Factor Authentication (MFA).
- XI. Implement Role-Based Access Control (RBAC).
- XII. Website is fully responsive and accessible on all devices and browser.
- XIII. Implement SEO best practices to improve the website's visibility in search results.

Guided Lab Exercises

The Lab Exercise Learning Paths provide structured, hands-on cloud training tailored to different certification and skill levels. These paths guide users through practical exercises with a combination of resources, ensuring an in-depth learning experience.

1. **Lab Learning Path Variations:**

- a. **Certification-Based Labs with Manual:** Includes a step-by-step lab guide aligned with certification objectives to help users practice relevant cloud concepts.
- b. **Certification-Based Labs with Manual & Video:** Provides both a detailed lab manual and an instructional video demonstrating the lab execution.
- c. **Comprehensive Certification-Based Labs:** A complete package that includes a lab manual, an instructional lab video, and a full training video covering theoretical and practical aspects.
- d. **Comprehensive Certification-Based Labs with Instructor Support:** Designed for learners who prefer guided labs with live instructor support for better understanding and troubleshooting.
- e. **Master Labs:** Advanced labs that consist of multiple sub-sections, allowing learners to complete a series of interlinked labs to achieve a specific goal, simulating real-world cloud projects.
- f. **Hands-on Cloud Challenge Labs:** Scenario-based labs that encourage learners to solve cloud challenges independently, reinforcing problem-solving skills.
- g. **Labs with Cloud Account Access:**
 - o When a user purchases a lab with cloud access, the admin manually creates lab account credentials and emails them to the user.
 - o The lab page clearly displays the available hours and time window (e.g., 9 AM – 6 PM) during which users can request access.

- A direct link (*email/ WhatsApp*) is provided, enabling users to send an access request when needed.
- h. **Learning Path Structure & Access Control:**
 - Displays all associated lab topics within each learning path.
 - Content remains locked until the user subscribes or purchases access (*only the headline and subtopic are visible before purchase*).
- i. **Search & Filtering Capabilities**
 - Advanced search functionality to quickly find relevant labs, courses, and resources.
 - Allows users to filter labs based on:
 - *Skill Level*: Beginner, Intermediate, Advanced.
 - *Format*: Video, Manual, Challenge Labs, Instructor-Led.
 - *Certification Type*: AWS, Azure, GCP, etc.
 - *Type*: Development, Administration, Architect, DevOps, etc.

2. Adding Content Key Features:

a. Customizable Text Formatting:

- **Font Customization**: Users/ Writer can modify font type, size, and colour for better readability.
- **Word Highlighting**: Enables text highlighting to emphasize important terms, instructions, or warnings.

b. Structured Content Support:

- **Organized Information Display**: Supports numbering, bullet points, and multi-level bullet styles for clear content structure.
- **Collapsible Sections**: Allows content to be expanded or collapsed for better navigation.

c. Code Integration & Accessibility:

- **Code Snippet Section**: Writers can add code blocks where learners can view and copy the code directly.
- **Syntax Highlighting**: Enhances readability and understanding of code samples.

d. Note Section:

- Show the Note section within the content with a bordered layout to emphasize key points. Example:

e. Multimedia Support:

- **Image Integration**: Instructors can insert relevant images, architecture diagrams, and visual aids.
- **Video Embedding**: Supports instructional videos to enhance the learning experience.

f. File Management & Downloads:

- **File Upload Section**: Instructors or admins can attach downloadable ZIP files containing code, scripts, or configuration files.
- **Pre-Configured Lab Files**: Learners can download and use pre-configured files to complete labs efficiently.

g. Version Control & Updates:

- **Automatic Versioning**: Each update is recorded with timestamps to track changes.
- **Content Update Logs**: Admins and content writers can view previous versions and modifications.

- **Learner-Friendly Display:** Users always see the latest, most updated version without confusion.
- h. **Multi-Device Compatibility:**
 - **Responsive Design:** Optimized for desktops, tablets, and mobile devices to ensure seamless access.
 - **Cross-Browser Support:** Works efficiently across different web browsers.
- i. **Badge and Certification System:**
 - **Automated Certification:** Generates badges or certificates based on the successful completion of a lab, skill, or learning path.
- j. **Lab Instruction Details:**
 - Each lab includes essential metadata for clarity and usability:
 - **Unique ID:** Each lab must be assigned a distinct identifier for easy search and allocation.
 - **Type:** Lab, Challenge, or Project-Based Exercise (displayed with icons).
 - **Estimated Completion Time:** Time required to finish the lab.
 - **Skill Level:** Categorized as Beginner, Intermediate, or Advanced.
 - **Cost:** Displays pricing based on country selection and subscription model.
 - **Points System:** Shows redeemable points that can be used towards a subscription plan.
- k. **Lab Integration:**
 - **Flexible Learning Paths:** A single lab can be linked to multiple learning tracks, making it applicable across different training modules.
- l. **Publishing & Drafting:**
 - **Draft Mode:** Labs can be saved as drafts for later updates before publishing.
 - **Publishing Controls:** Admins can publish labs to make them live on the website.

3. User Registration

- Users can register using their email address.
- OTP authentication at the time of Registration (integrated with an SMS gateway) for enhanced security.
- Mandatory Two-Factor Authentication (2FA) using authenticator apps, SMS codes, or email verification.
- Alternative login options via Google, LinkedIn, or other social media platforms.
- Secure password recovery and reset functionality (*Periodic reset must*).
- Users must complete their profile setup, including personal details and a profile photograph (*For Corporate users, must add employee Id*).
- Option to update profile information, including professional background and skills.
- Note: User credentials (passwords) must be securely protected to ensure they are not visible or accessible in any form, preventing unauthorized access and data breaches.

4. User Dashboard

The User Dashboard provides a personalized and interactive learning experience, helping users track their progress, manage enrolled labs, and engage with course content effectively.

- **Selection & Access:**
 - **Learning Path Selection:** Upon login, users must select a preferred learning path to tailor their experience.
 - **Advanced Search & Filtering:** Users can search, and filter labs based on skill level (Beginner, Intermediate, Advanced), format (Video, Manual, Challenge Labs), certification type, and cloud platform (AWS, Azure, GCP, Oracle, etc.).
 - **Subscription-Based Access:** Users can only view the details of content they have subscribed to.
- **Progress Tracking & Engagement:**
 - **Visual Progress Tracker:** Displays completed, ongoing, and pending labs within the selected learning path.
 - **Achievements & Badges:** Users earn completion badges, certificates, or skill-based recognitions upon successfully finishing labs.
 - **Lab Status Indicators:** Clearly marked statuses such as "Not Started," "In Progress," "Completed," and "Expired" with countdown timers for time-based labs.
 - **Automatic & Manual Tracking:**
 - Labs close automatically upon completion.
 - Users can manually start, pause, or close labs based on their pace with session tracking.
 - Option to abandon a lab if necessary.
 - Auto close on time expiry for timed labs.
- **Interactive Features for Enhanced Learning:**
 - **Notes & Annotations:** A built-in Notes Section where learners can take personal notes while working on labs.
 - **Discussion Section:**
 - Interactive discussion for instructor-led training subscribers to engage directly with trainers (*Enable if user subscribe to the instructor led lab*).
 - **Feedback and Rating System:**
 - Allow learners to provide feedback and rate courses and instructors (view by admin only).
 - **Bookmarking & Saved Content:** Users can bookmark labs for quick access and revisit important materials.
 - **Upcoming Classes & Events:** A dedicated section for tracking scheduled live training, webinars, and expert talks.
- **Achievements & Milestones:**
 - **Certification & Training Milestones:** Displays earned certifications, completed courses, and skill-based achievements.

5. Role Types

The platform supports multiple role-based access controls (RBAC) to ensure secure and efficient management of content, subscriptions, and user interactions.

- **Admin Role (Super Admin)**
 - Full access to all system functionalities.
 - Can manage users, roles, subscriptions, content, payments, and platform settings.
 - Has ultimate control over approvals, reports, and system configurations.
- **Approval Role (Subscription Manager)**
 - Responsible for reviewing and approving module subscriptions.
 - Can manage user enrolments, process approvals/ rejections, and track subscription statuses.
 - May oversee discount applications, free coupons, and refunds if applicable.
- **Writer Role (Content Creator & Manager)**
 - Can create, edit, and manage lab content (articles, manuals, videos, challenges).
 - Can upload resources (code files, images, documentation, etc.).
 - May have limited publishing rights, requiring admin approval before content goes live.
 - Can track versioning and updates to maintain content integrity.
- **User Role (Learner/Subscriber)**
 - Can browse and access only subscribed content.
 - Can participate in hands-on labs, quizzes, and training sessions.
 - Has the ability to bookmark, track progress, and earn badges or certifications.
 - Can engage in discussions (if enrolled in instructor-led training).
- **Support Role (Customer Assistance):**
 - Handles technical support, user queries, and troubleshooting issues related to labs and platform access.

6. Certification Test Series

Develop a test series tailored for specific certifications.

- Showcase a dedicated homepage for the Test Series.
- Each test will feature multiple-choice questions, with:
 - Options to select one or multiple correct answers.
 - Options available for drag-and-drop selection.
- **Learning Mode:**
 - Users can view the correct answer along with an explanation immediately after each question.
 - No time restrictions.
- **Exam Mode:**
 - Users can complete the test within a set time limit and review the answers only after submission.
 - Offer users detailed performance reports with areas of strength and weakness.
 - Option to receive a certification upon successfully passing the test.

- Users can purchase the test series as per their requirements community.
- Note: Learning Mode and Exam Mode can be offered separately or bundled as a single product (similar to the feature available in Labs).

7. Cloud Account Services

Option to purchase cloud-based services with flexible duration plans.

- A dedicated page section highlighting available cloud account services.
- Purchase cloud accounts pre-configured for specific certifications, including predefined services and durations.
- Option to select customized cloud services and duration, with an automated inquiry email for personalized requests.

8. Instructor Resources

Empower trainers with access to high-quality course materials and continuous updates.

- Showcase a dedicated homepage for the instructor resources.
- Download trainer slides upon purchase, available per module or certification.
- Opt-in for long-term access to receive updated content aligned with changes.
- Automated email notifications for version upgrades, ensuring trainers stay up to date.

9. Instructor-Led Training

Participate in scheduled live training sessions.

- Showcase a dedicated homepage for the instructor led training.
- View a detailed table of upcoming training sessions, including:
 - Training name
 - Registration and purchase link
 - Duration
 - Mode of delivery type (Online/Offline)
 - Sending Training link and other communication will be handle manually

10. On-Demand Video Training

Pre-recorded video sessions for flexible learning.

- Showcase a dedicated homepage for the On-Demand Video.
- Option to purchase based on the module.

11. Career Assistance

Resume-building support and interview preparation.

- Dedicated homepage showcasing additional career services and upgrade options.
- Offer mock interviews with industry experts to help users prepare.
- Purchase premium career assistance packages.

12. Exam Voucher

- Dedicated homepage showcasing exam voucher details.
- Option to buy certification exam vouchers.

13.Free Webinars & Live Events & Expert Talks

- Dedicated homepage showcasing upcoming free sessions, industry events, expert discussions.
 - Option to register.
 - Sending link and other communication will be handled manually.

14.Others

a. Integration

- ✚ The homepage will showcase relevant content.
- ✚ Users can register on the website.
- ✚ View is limited to the Lab and Test Series sections.
- ✚ Users can be grouped together by Admin.
- ✚ Admins can assign a learning path, specific guided labs, or practice tests to user groups.
- ✚ Users can only access the guided labs or practice tests assigned to them.

b. Subscription & Discounts

Purchase training modules with exclusive discounts and free coupons.

- ✚ Purchase individual training modules as per your learning needs.
- ✚ Buy training points to unlock and select multiple modules of your choice.
- ✚ Apply discount coupons to avail special offers on purchases.
- ✚ Payment gateway integration for both national and international transactions.

c. Referral Program

Implement a referral program to incentivize users to bring in new customers with additional discounts.

- ✚ Enable external partners to promote courses and earn commissions.
- ✚ Special plans for companies to train their employees.

d. Skill Assessment Quiz

An adaptive, interactive quiz that guides users through a series of targeted questions to determine their career interests and skill goals.

Based on each response, the system dynamically presents follow-up options (*e.g., selecting Development may lead to choices like C, Python, or Java*).

At the end of the quiz, users receive personalized recommendations for the most suitable training courses or labs aligned with their chosen path.

e. Security Requirement:

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- ✚ Enable OTP-based registration and enforce Multi-Factor Authentication (MFA).
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- ✚ Website is fully responsive and accessible on all devices and browser.
- ✚ Implement SEO best practices to improve the website's visibility in search results.

f. Integration with Second Website:

The backend will integrate seamlessly with a secondary website (hosted under a different DNS) that serves as a simplified version of the primary platform. This site will exclusively display Lab and Certification/Test Series modules.

- ✚ The homepage will showcase relevant content.
- ✚ Users can register on the website.
- ✚ View is limited to the Lab and Test Series sections.
- ✚ Users can be grouped together by Admin.
- ✚ Admins can assign a learning path, specific guided labs, or practice tests to user groups.
- ✚ Users can only access the guided labs or practice tests assigned to them.

g. Enable/ Disable

Add an option to enable or disable the following pages on the homepage:

- ✚ Free Webinars & Training
- ✚ Live Events & Expert Talks
- ✚ Instructor Resources
- ✚ Instructor-Led Training
- ✚ On-Demand Video Training
- ✚ Career Assistance
- ✚ Exam Voucher.
- ✚ Note: This ensures these pages are displayed on the homepage only when their content is ready.